

Mining the Stream of News for City Areas Profiling: a Case Study for the City of Rome

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Abstract—Tracking and profiling changes in the occurrence of notable events in a city, in terms of what happens in the different areas and how possible changes are perceived, is an important issue in the context of smart cities: in fact, it may be helpful in developing applications to help administrations and citizens alike. In this paper, we propose an approach to provide time-sensitive snapshots of events within the different areas of a city, and the city as a whole. To probe inside neighborhoods and communities, we propose to use articles in online newspapers, as they represent an accessible source of information on what notable events actually happen, and on the most relevant topics at a given moment in time. We adopt an approach to group up articles by means of clustering, and to automatically assign labels to clusters by analyzing their content. The outcomes of this procedure, repeated along a certain timespan, are able to describe the temporal evolution of notable events in specific city areas. In this paper we show the effectiveness of the proposed methodology by reporting a case study for the city of Rome, over an investigation span of few years, which includes also the Covid-19 pandemic period.

Index Terms—City Areas Profiling, Online News Clustering, NLP, Text Mining, Streaming Data, Smart Cities

I. INTRODUCTION

In the last few years, research on smart cities has experienced a substantial growth due to the pervasive presence of *IoT* enabled devices in the urban environment, and to the ever growing need of clear and up-to-date information on the most important aspects of the city life. On the one hand, this is helpful for regulators and administrations, as they can better analyze how the city evolves and how to better control and direct such changes. On the other hand, citizens can benefit from knowing what happens around them and how to fulfil their specific needs in the city. In fact, if we look at the literature, the core idea of a smart city is that of improving the quality of life via the smart use of technology [1]. In this context, the aspect of profiling city areas plays an important role [2]. The main goal of profiling is making generalizations about certain groups of items based on shared aspects or interests, that can be identified in several different ways. Such generalization can be subsequently exploited for example to perform the categorization of new items. Within the context of smart cities, profiling techniques have been

exploited to pursue several different goals and consider a wide array of data. For example, works have been proposed for the profiling of areas based on urban activities and the various types of points-of-interest in neighborhoods, as well as the interactions between citizens [3], [4]. Large scale data analysis and social media posts have been exploited successfully in the *CityPulse* framework [5] to provide information to both local governments and citizens. In some previous works, we focused on profiling city areas in the context of smart city applications. Specifically, we proposed a framework that uses various on line sources (e.g. traffic flows, house pricing, points of interests etc.) to automatically profile city areas in [6]. Further, in [7] and [8] the framework was extended to enable the exploitation of newspaper articles as well. We proposed a method to label the articles as pertaining to a *macro-category* (e.g. TRAFFIC, CRIMES, URBAN DECAY). It was developed a method to automatically identify such categories from tags of newspaper articles, considering the semantic similarity across tags, expressed through pre-trained word embeddings. Finally, the geo-localization of article news was used to profile different urban areas.

Two main drawbacks from previous approaches can be identified. First, news categorization was based on a supervised learning approach. This may be impractical in real-world applications as it requires human intervention and labelling. Moreover, the macro-categories were generated semi-automatically, i.e. category labels were manually selected. Second, we did not provide a semantic-aware representation of the articles themselves, which may instead enable the identification of similar newspaper articles regardless of their tags, and in turn improve the ability to discover new groups.

In order to address these shortcomings, we propose to exploit vector representations of words and sentences, paired with unsupervised data mining techniques. On the one hand, we evaluate the use of state-of-the-art neural language models to represent the text of the news articles. In fact, Natural Language Processing (NLP) techniques focused on the semantic representation of words and texts have made great strides in the last ten years. This success was achieved especially thanks to neural models that learn unsupervised representations of words and sequences by means of their distribution. Differently from the earliest works based on neural learning algorithms, such as *word2vec* [9] and *fastText* [10], which focused on learning the

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representation of single words by learning to predict their context in usage, or vice versa, the latest language models, based on the Transformer architecture and the attention mechanism [11], can be considered as a breakthrough in NLP research. They are able to leverage the pre-training and fine-tuning learning paradigm, and can provide representations for entire sequences at once without the need for recurrent architectures. On the other hand, we propose the adoption of density-based streaming clustering algorithms, in order to better model the evolution of news and thus provide more up-to-date profiles for the city. In order to evaluate the approach, we describe a case study on the city of Rome. In this experiment, we track the evolution of news referring to places in the whole urban territory, and we consider specific areas and neighborhoods by looking at cells in a bidimensional grid superimposed over the city.

The main novelty of this paper compared to related previous works [7], [8] consists in the adoption of a fully unsupervised pipeline for text mining: to the best of our knowledge, this is the first time that semantic-aware text representation and density-based streaming clustering are combined in a news-based framework for city areas profiling.

This paper is organized as follows. In Sec. II we provide an overview of the proposed approach. Section III shows the obtained results on the case study for the city of Rome, and Sec. IV provides a discussion on such results. Finally, Sec. V draws some conclusions.

II. PROPOSED APPROACH

Our main goal is to provide an effective and efficient method to track changes happening in specific areas of the city, or the city as a whole, in specific periods of time. Three key issues must be addressed in the proposal of a proper solution. First, the selection of the information source to best represent the significant events occurring within the area of interest. The relative data should be obtained as a continuous stream to support an up-to-date system, and should contain both general knowledge about the city and specific aspects of interest at a given time. Second, we need a representation method able to provide a representation of the semantics of data, and suited to machine learning algorithms. Third, we need an effective algorithm/model to tell apart the various aspects of interest, deriving this ability directly from data with no (or minimal) supervision, and able to deal with continuous inputs as well. Finally, a crucial role is played by the visualization strategy of the results.

The proposed approach is represented in Figure 1. Specifically, we first consider and collect data and metadata from online newspapers for specific time-frames. Then, we represent them in a *distributed vector space* leveraging a state-of-the-art neural language model. Further, in order to obtain groups of news, we propose to perform dimensionality reduction on the distributed representation and cluster the pieces of news. Finally, we automatically label the clusters with a set of tags, thus enabling the evaluation and profiling of city areas at a given time.

A. Data source and data collection

As for the data source, we chose to exploit articles from local online newspapers. We can argue that they can effectively incorporate a large number of information and cover a wide array of topics, that can in turn better describe a specific moment in time without the need of focusing on specific aspects of life in the city. To corroborate this, previous research has in fact shown us that newspaper articles, and tags contained in them, are a viable option when building categorization systems to profile city areas [8].

Typically, such news articles already provide information concerning the three most important aspects for this research, namely the time they are published, some form of geo-localization (e.g. the name of the neighborhood or roads), and the content of the actual news. In addition to this, often they also include a list of *tags* that describe the general topic of the news (e.g. THEFT, ROAD ACCIDENTS, CORONAVIRUS). They can be considered as describing in some sense the category the news pertain to, and thus can be exploited for providing some form of labelling to groups of them. In fact, we exploit them to automatically label clusters of news for a given time period.

As our primary data source, we chose to exploit one of the Today¹ websites. These websites, owned by the *CityNews* company, provide free online news for more than 50 Italian cities. The actual data obtained for each article includes a *title*, a *summary*, and the actual *body*. In addition, extracted metadata that are relevant for our analysis are the *time of publication*, the optional list of *tags* provided by the author of the article, and an optional location, for which we found the actual geographical coordinates through Nominatim², a search engine for OpenStreetMap data. The search engine enables search either by looking up names and addresses or via geographic coordinates.

The data and metadata collected are exemplified in Table I.

TABLE I
RAW DATA EXTRACTED FROM NEWS ARTICLES

Type	Sample value
URL	https://www.romatoday.it/cronaca/coronavirus-mascherina-abusive-quanto-costano.html
Time Stamp	2020-02-27 10:03:31
Title	Coronavirus, 5 euro per una mascherina: venditore abusivo finisce nei guai (<i>Coronavirus, 5 euros for a mask: abusive seller ends up in trouble</i>)
List of Tags	Coronavirus
Location	Centro Storico, Via Quattro Novembre
Summary	Nei confronti dell'ambulante fuorilegge è scattata una sanzione amministrativa di 5.000 euro per la violazione del Regolamento di Polizia Urbana (<i>An administrative fine of 5,000 euros was imposed on the outlawed peddler for the violation of the Urban Police Regulations</i>)
Body	La paura scatenata dalla diffusione del coronavirus ... (<i>The fear triggered by the spread of the coronavirus ...</i>)

¹<https://www.today.it>

²<https://nominatim.openstreetmap.org>

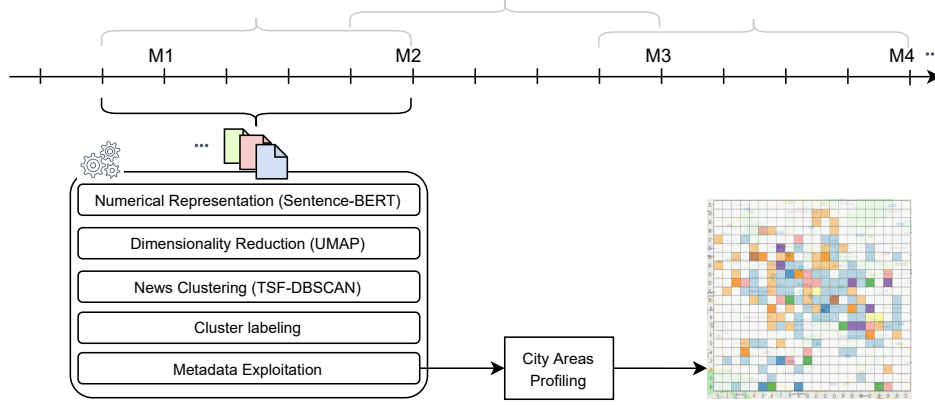


Fig. 1. The pipeline for the proposed system.

B. Distributed representations of newspaper articles

The inherently unstructured nature of texts requires some form of machine-readable representation that can encode their meaning or part of it in order to be able to process it via data mining algorithms such as clustering.

In the last few years, neural language models have imposed themselves as the de-facto standard for most NLP tasks, including feature extraction for words and sequences of texts. These features are aimed at providing a distributed representation of the meaning of words and sequences, and enable comparisons with similarity metrics. Transformer language models such as BERT [12] have obtained state-of-the-art performances in most downstream NLP tasks. However, as for the feature extraction, authors in [12] clearly state that BERT-based representation of the entire sequence, differently from the token-level representations, are not to be considered semantically meaningful. Thus, a popular approach to address this challenge was proposed in [13]. Authors propose the use of siamese BERT networks to learn a model tailored to address the semantic similarity of sentences. Specifically, the model consists actually of two identical BERT models that jointly learn the representations of pairs of sequences in order to obtain vectors that better encode the similarity between the two sequences. Starting from a pre-trained transformer model, authors fine tune it on (i) tasks of semantic textual similarity [14], where two sequences must be assigned with a similarity score, and (ii) on sequence pair classification [15], [16], where two sequences must be classified for example as entailing each other. The resulting model, named Sentence-BERT [13], is able to provide vector representations of sentences that can be compared for example via cosine similarity.

For our specific problem, we intend to obtain representations for Italian news, and thus for long sequences. To this aim, we leverage the `xlm-r-bert-base-nli-stsb-mean-tokens` multi-lingual Sentence-BERT pre-trained model. Since for very long input sequences the resulting representation may be less meaningful for our goal, we chose to encode only the title and the summary of each article. We feed the concatenation of title and summary to the model and obtain a

768-dimensional embedding for each article.

C. Profiling via clustering

The numerical representation of textual data enables a downstream analysis aimed at extracting knowledge from the news streams. The goal of such analysis is to cluster documents in order to get a fine-grained picture of what happens in the city, both on a temporal and a geospatial scale. Since the distributed representation of the documents encodes their semantics, objects belonging to the same cluster are expected to be associated with the same or similar topics.

As for the clustering algorithm, we revisit the recently proposed TSF-DBSCAN [17]. It consists in a two stage, online-offline, density-based clustering algorithm for data stream: during the online stage, objects are organized in adjacency lists; during the periodically triggered offline stage, a forgetting mechanism is applied to discard outdated objects and an adaptation of the well-known DBSCAN algorithm is used to obtain the output partition. TSF-DBSCAN leverages concepts from fuzzy set theory: it relaxes the constrain on the neighborhood size (parameter ϵ) thus decoupling the identification of core objects (based on a *tighter* threshold ϵ_{min}) and the membership assessment of border objects (based on a *looser* threshold ϵ_{max}). As a result, TSF-DBSCAN can capture clusters with fuzzy overlapping borders. We argue that this property may turn out to be particularly useful in a context such as document clustering, where the boundaries between clusters representing topics may not necessarily be sharp.

However, document clustering approaches must cope with a problem known as the curse of dimensionality: with the increase of the number of dimensions, all the instances appear far to each other, and consequently no high density region can be identified. The problem affects most clustering algorithms, including TSF-DBSCAN. For this reason, we have revisited the original TSF-DBSCAN: the online stage is simply in charge of storing the numerical representation of articles; when a chunk is collected (i.e. offline stage is triggered), we adopt UMAP [18] as a pre-processing step prior to cluster analysis in order to project the article-level embeddings obtained by Sentence-BERT ($dim = 768$) into a low dimensional space

($dim = 5$). We empirically tuned the hyper-parameters of UMAP based on literature suggestions [19] and our own results.

The offline stage of TSF-DBSCAN is triggered with a period of one month, and the forgetting mechanism is tuned so that there is an overlap of one week between two consecutive clustering evaluations. In other words, every reclustering step considers items gathered in the preceding five weeks, approximately. To enhance the flexibility of the algorithm in identifying clusters in an evolving setting, the parameters ε_{min} and ε_{max} have been automatically tuned at each reclustering stage following the unsupervised approach presented in [20]. To sum up, the novel tweaks applied to TSF-DBSCAN let us handle high dimensional data by exploiting a dimensionality reduction transformation, and tame the evolving nature of the news stream as well, through the adaptive setting of the distance threshold parameters.

D. Automatic cluster labelling

In order to interpret the partition discovered by TSF-DBSCAN, we implement an unsupervised clustering labeling module: each cluster is annotated with the *tag* that best describes the relative collection of news. Specifically, for each cluster we evaluate the ten most frequent tags that appear in the metadata of its news articles; then, we obtain a numerical representation of each tag based on word embedding, specifically with *fastText* [10]. Finally, we label each cluster as follows: we evaluate the average of the tags embeddings, each one weighted according to its frequency, and label the cluster with the tag with the minimum cosine distance from such average vector. In addition, we record the most frequent tags in the documents of the cluster for each specific area of the city.

As the last step of our pipeline, we exploit the *location* information available in the news articles metadata to carry out a geospatial analysis and identify what clusters are prevalent in the different areas of the city.

Each algorithmic component of the system requires setting up specific parameters: we adopted the parameterisation suggested in the specialised literature. A summary is reported in Table II. Notably, the *cosine* distance adopted in UMAP is widely exploited in NLP tasks to measure the distance between text embeddings. The limited size of the local neighborhood used to learn the manifold structure of the data ($n_neighbors$) should allow us to obtain a good balance between targeting global and local structures in the reduced space, whereas the number of components has been set to 5 to enable the downstream clustering procedure.

III. CASE STUDY: ROME TROUGH ROMATODAY

In order to evaluate our approach, we considered a case study for the city of Rome. We collected a dataset of news articles from *RomaToday* in the time span from 01-01-2019 to 28-02-2021. In such context, we argue that the horizon of one month for the reclustering stage allows, on the one hand, to collect enough data points and, on the other hand to grasp significant fluctuations over time. Overall, we collected

TABLE II
MAIN PARAMETERS ADOPTED IN OUR SYSTEM

SentenceBERT	$model = xlm-r-bert-base-nli-stsb-mean-tokens$
UMAP	$n_components = 5, n_neighbors = 10,$ $metric = cosine$
TSF-DBSCAN	$(\varepsilon_{min}, \varepsilon_{max})$: Automatically inferred as in [20] $MinPts = 5, metric = euclidean$

30,239 news articles but only around 15,200 articles can be geo-localised. In addition to this, around 3,000 articles do not have any tag associated. As for the tags, we identified 5,787 distinct ones. For the labelling of clusters, we filtered out the tag list based on frequency and through a manual evaluation, in order to keep only the most relevant and informative tags. For example, tags such as “Rome”, that are very frequent but uninformative, were removed during the manual evaluation. We finally obtained a list of 311 valid tags. Second, we obtained the distributed representation of articles with the Sentence-BERT model. As explained in Section II-B, we consider only the title and summary of the article in order to obtain a more reliable representation. Finally, we apply the TSF-DBSCAN clustering algorithm to periodically obtain a partition of the most recent articles. At each evaluation, the discovered clusters are labelled according to the method proposed in Sec. II-D.

For the sake of brevity, we propose here to evaluate two aspects of the results. First, we show how the proposed system is able to identify and geo-locate what happens in the city via clusters of news articles. Figure 3 provides a snapshot of the whole city of Rome for the time frame ranging from 24-04-2020 to 01-06-2020. The superimposed grid divides the city into smaller square cells, 1km wide. The color of each cell corresponds to the most representative cluster in the cell in that specific time frame. moreover, we also report the three most frequent tags for each cluster. As expected, due to the Covid-19 pandemic during these months, and the lockdown actually in effect for the months of April and early May, most areas are actually labelled with the CORONAVIRUS tag. We can also see how such tag appears in other clusters, demonstrating that the pandemic has pervaded different aspects of the city’s life: from this perspective, the fuzzy clustering approach may represent an added value for the extraction of knowledge from the online news if compared to the mere use of the available tags. Interestingly, we see that many THEFTS are found near the city centre, while cluster labels as ROBBERY are found in more residential areas.

Second, we provide an evaluation of the evolution of clusters for specific cells across the whole time span. This can be considered as the main outcome, as it enables a finer-grained analysis of the areas of the city. We show here an example for a cell that includes the Rome Termini train station, that is one of the most important travel hubs of the Italian Capital, and Piazza Vittorio Emanuele II. Figure 2 shows the obtained results for this cell. For readability, we only report the most

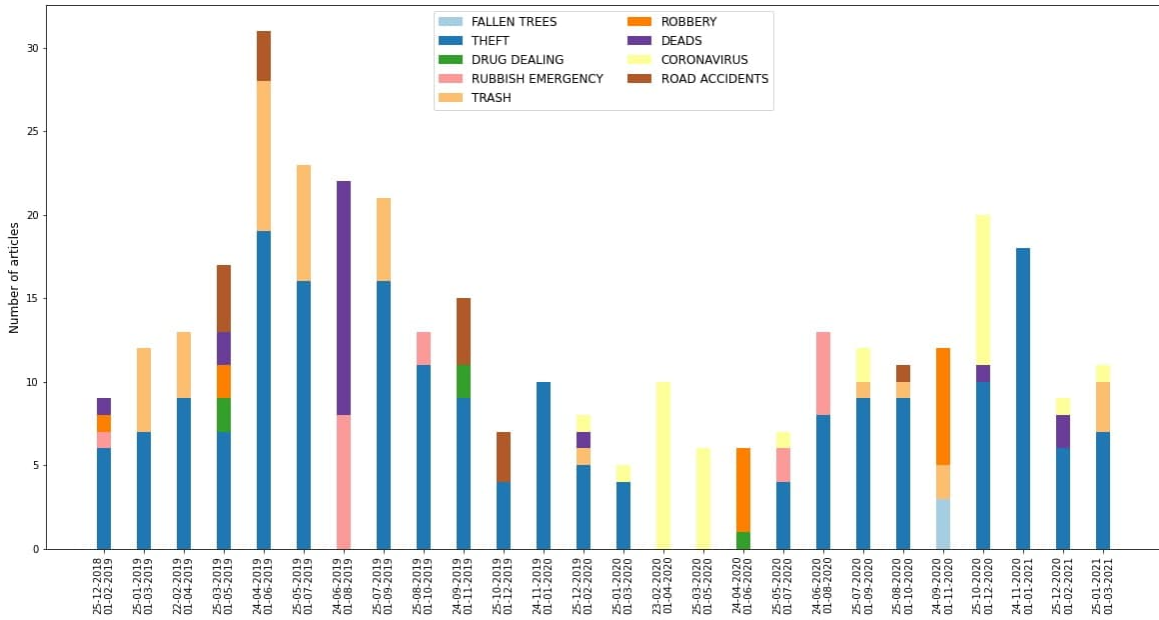


Fig. 2. Cluster history for *Cell 11_10*: the area of Rome Termini train station and Piazza Vittorio Emanuele II.

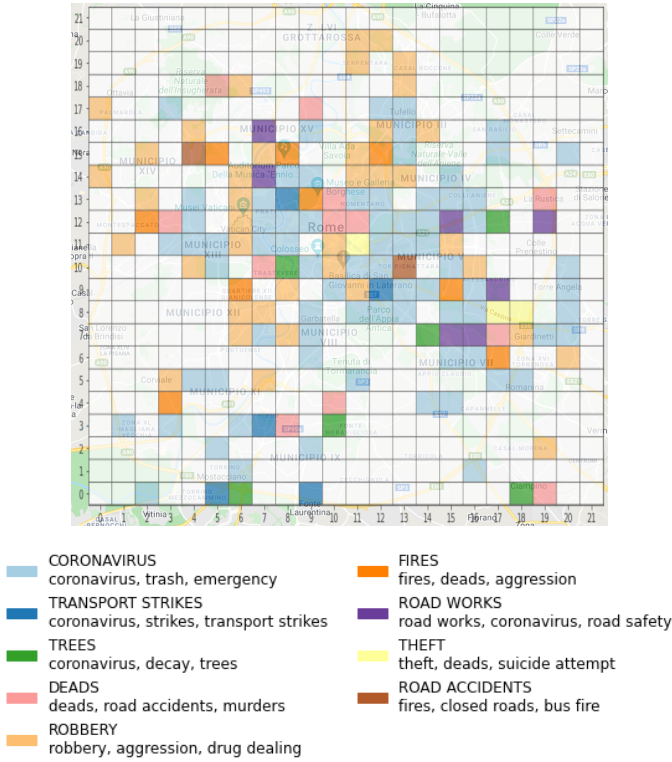


Fig. 3. Captured clusters for the period 2020-04-24 - 2020-06-01.

frequent clusters in the overall time frame considered in the plot. For each time slot, we represent the number of points for a given cluster in the specific time window. The figure provides the evolution of the clusters over time. It is interesting to observe how THEFT are almost always represented in each

time window, with the exception of the months corresponding with the nation-wide lockdown for the Covid-19 pandemic. In that case, only CORONAVIRUS-related clusters are represented. In addition, it appears that the overall volume of news for the area in 2020 decreases, probably due to the pandemic and specifically to the lockdown as well, which have reduced the number of events and the mobility of people significantly. We see a spike in Covid-related clusters also in November 2020, when Italy went through a second, less restrictive lockdown. In this case, it looks like crimes were not affected. It is also interesting to notice the spike in DEADS for July 2019. By looking at the news, we noticed that in July 2019, more deaths by traffic accidents than usual were actually reported, thus proving that the system can effectively identify relevant events and widely discussed topics in news.

IV. DISCUSSION

The case study on Rome can give us some insights on the effectiveness of the proposed method. First, we saw that employing an unsupervised strategy based on streaming clustering algorithms enabled us to effectively model how news evolve over time without prior knowledge of the main topics, and in turn profile their relevance both city-wide and on a smaller scale. The examples provided have in fact shown us that the method is rather effective in identifying the most important aspects reported in news, how they unfold over time, and how they relate to each other. For example, we were able to see the effect of the Covid-19 pandemic on the level of criminality for a given part of the city.

Second, we evaluated the effectiveness of neural language models to represent sequences of texts for this task. If we consider textual data as our primary source for modelling smart cities, the availability of such models is an added value

also from an application perspective, as they provide pre-trained resources that do not require further computation.

Third, we propose a pipeline for data representation and modelling that is able to effectively address changes in the data over time without further computation, and is able to provide to end users a better understanding of the evolution of important topics and events related to a specific city. For example, we can see that the problem of RUBBISH is recurring, and often relevant for the city of Rome. We can argue that this may enable the implementation of strategies to better address such problems, both at a neighborhood level and city-wide.

Going forward, we must acknowledge some limitations. First, the pre-trained nature of the text representation method used in the proposed approach, even if useful in real-world applications, may yield poorer results with respect to using language models tailored to the specific textual genre and topics. Second, the evolving nature of the news stream should be further investigated: in the streaming setting, clusters may evolve over time by splitting, merging, or disappearing: a mechanism for the rigorous tracking of clusters is desirable in order to effectively follow their trends. In addition, the labelling strategy could be improved in order to reflect the relationship between news and clusters as well, rather than focusing only on the main topic of a cluster.

V. CONCLUSIONS

In this paper, we have proposed a method to extract topic profiles from news articles, and to use them for in turn profiling city areas. With respect to our previous work, that focused on a set of pre-determined categories, the proposed approach has allowed us to automatically identify relevant topics based on the tags, and to track reporting over time, in terms of which clusters appear in a given neighborhood.

The proposed approach considers online newspaper articles as data source, in order to obtain reliable and continuously updated information regarding a specific city. State-of-the-art neural language models have been used to encode the semantic information of the articles in fixed-length vectors. Such vectors have shown to reliably represent the information and enable the comparison between news articles via vector similarity metrics. The TSF-DBSCAN algorithm has been applied to the stream of news to periodically obtain a partition of the documents. The resulting clusters have been automatically labelled by considering the tags of the articles. Finally, the clusters have been exploited for providing an overall map of the city in a specific time window, and to plot the evolution of the profiles over time for specific areas of the city.

We can argue that the proposed approach, despite the need for a more thorough evaluation through different case studies, as well as in more controllable environments, is nonetheless effective in providing a snapshot of what are the most important and discussed topics at a given time in a given city or area of it, thanks to its unsupervised nature. This may prove to be rather useful for evaluating how city areas change and evolve over time, both in normal and crisis times such as a global

pandemic, and help in making predictions and taking more informed decisions that concern the well being of citizens.

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